Parallel Lines

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Parallel lines

What are parallel lines?

- **Parallel lines** are **straight lines** with the **same gradient**
  - Two parallel lines will **never meet**
    - They just stay side-by-side forever
- The **equation** of the line **parallel** to $y = mx + c$ is $\boldsymbol{y = mx + d}$
  - $y = 2x + 1$ and $y = 2x + 5$ are parallel
  - $y = 2x + 1$ and $y = 3x + 1$ are **not** parallel

How do I find the equation of a parallel line?

- For example, to find the **equation** of the line **parallel** to $y = 2x + 1$ which **passes through** the **point** (3, 14)
  - write the parallel line as $y = 2x + d$
    - using the same gradient of 2
  - **substitute** $x = 3$ and $y = 14$ into this equation
    - $14 = 2 \times 3 + d$
    - $14 = 6 + d$
  - **solve** to **find** $\boldsymbol{d}$
    - $d = 8$
  - The equation is $y = 2x + 8$

Help me understand

Worked Example

Find the equation of the line that passes through the point (2, 1), which is parallel to the straight line $y = 3x + 7$.

Answer:

Parallel means the gradient will be the same
Use $y = mx + d$ where $m = 3$

$$y = 3x + d$$

Substitute in $x = 2$ and $y = 1$

$$1 = 3 \times 2 + d$$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Solve the equation to find d (by subtracting 6 from both sides)

$$-5 = d$$

Write out the final answer in the form $y = mx + d$

$$y = 3x - 5$$

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